Sahasra Ranjan

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Education

Indian Institute of Technology Bombay — Mumbai, India
B.Tech in Computer Science and Engineering | GPA: **9.05**/10 | Honors GPA: **9.75**/10 — *Aug 2019 - May 2023*

Selected honors

- **ACL 2023 Outstanding Paper Award**, co-first author; awarded to less than 1% of accepted papers *2023*
- **Research Excellence Award**, best Bachelor's Thesis among graduating CS students, IIT Bombay *2023*

Industry experience

Socratic AI Inc — Aug 2025 - Present
Founding AI Engineer (2nd engineer) — *Sunnyvale, CA*

- Designed and built a synthetic-company simulation environment for **long-horizon evaluation of AI agents**, in which **21 simulated employees** generate email, chat, calendar, and spreadsheet activity calibrated to production telemetry; agents are scored through checkpoint probes and LLM-based graders.
- Developed evaluation infrastructure for real-time and workflow agents, including an LLM-as-judge and prompt-injection test harness, a tool-calling and latency benchmark across real-time models, and regression replay over **3,300+ captured production traces**.
- Architected a unified **real-time speech-to-speech inference layer** across four model providers, handling streaming audio, turn-taking, voice activity detection, and barge-in; supported calls longer than Gemini Live's 15-minute session limit through context compression and session rollover.
- Built a **persistent memory system** for AI agents, comprising an append-only transcript store, source-grounded fact extraction, and two-stage nightly consolidation with rollback safeguards; implemented cost-aware model routing under per-user and organization-wide spend limits.
- Led engineering, reporting to the founder, for one of the company's two enterprise deployments under **multi-year, multi-million-dollar contracts**; introduced provenance validation for AI-generated requirements documents.
- Automated enterprise contact-center migrations through authenticated extraction of 13 configuration categories across region-specific APIs; shipped real-time SDKs in Python, TypeScript, and Swift and an ERP connector that **reduced migrations from months to hours**.

Rubrik Inc — Summer 2022 (Intern), 2023 - 2024 (Full-time)
Software Engineer, Data Life Cycle Team — *Bengaluru, India*

- Designed garbage collection for snapshot metadata, including reference tracking, a mark-and-sweep lifecycle, and gRPC APIs for detecting and removing orphaned entries across distributed storage nodes.
- Extended retention lock and legal hold to cloud-native workloads, enforcing immutability for **petabyte-scale backups**; implemented asynchronous SLA assignment using an event-driven poller across thousands of protected objects.

Entrepreneurship

Mathkraft — 2024 - 2025
Co-founder, AI Math Tutoring Platform for K-8 — *San Francisco, CA*

- Co-founded an AI math tutoring platform that embeds problems in student-created stories, leading a team of 4 engineers and 6 tutors; piloted with **150+ students** at **95%+ retention** over six months.

Research experience

Multimodal Models for Protein-Small Molecule Interactions
Research Intern and Collaborator | Prof. Martin J. Lercher, Alexander Kroll — *HHU Düsseldorf, Germany*

- Developed ESP, a general enzyme-substrate prediction model that jointly encodes protein sequences (fine-tuned ESM-1b) and molecular fingerprints; curated 70k BRENDA pairs with negative sampling of plausible non-substrates, and end-to-end fine-tuning raised accuracy from **82% to 91%**; deployed as a public web server.
- Designed ProSmith, a multimodal transformer with bidirectional cross-attention between protein and molecule representations and contrastive pre-training on 500k BindingDB pairs; reached **94.2% accuracy** on kinase inhibition and enzyme-substrate benchmarks (30% error reduction over the prior state of the art), serving **50k+ predictions** at deepmolecules.org.

- Developed and delivered a substrate-prediction model to a **global pharmaceutical partner**, predicting which drug-metabolizing enzymes and transporters interact with unseen drug molecules; extended a contrastive protein-molecule model (ESM-2, MolFormer) into a six-model ensemble with gradient boosting, reaching **MCC 0.53** and **AUC 0.88** on molecule-held-out splits.
- Designed the evaluation methodology, including molecule-held-out and leakage-controlled (DataSAIL) splits, fixed-ratio negative sampling, and metrics restricted to experimentally confirmed interactions; data audits identified conflicting labels and inflated benchmark performance in public CYP datasets.
- Added uncertainty estimation that reduced expected calibration error from **0.123 to 0.020** and accepts 40% of predictions at an 85% accuracy target; prototyped an LLM-agent pipeline for extracting enzyme-substrate interactions from full-text literature.

B.Tech Thesis Research May 2022 - May 2023
Department of Computer Science and Engineering *IIT Bombay*
Pretraining for Code-Switched NLP | *Advisor: Prof. Preethi Jyothi, CSALT Lab* *May 2022 - Dec 2022*

- Showed that standard masked language modeling ignores language boundaries, leading multilingual models (mBERT, XLM-R) to learn spurious monolingual correlations on code-switched text.
- Designed boundary-aware masking strategies that mask tokens at language switch-points: **Switch-MLM**, which uses language-ID tags, and **Freq-MLM**, which infers token language from monolingual corpus frequencies and needs no tagged data.
- Modified the encoder architecture with residual connections from an intermediate layer and an auxiliary switch-point prediction loss; together these improved NER and sentiment analysis by **5-7 F1** across Hindi-English, Spanish-English, Tamil-English, and Malayalam-English benchmarks.

Gradient Modification for Continual Learning | *Advisors: Profs. Preethi Jyothi, Manoj Prabhakaran* *Jan 2023 - May 2023*

- Developed **CommitteeSGD**, a continual-learning optimizer that keeps a committee of prior examples and damps or projects out the components of each gradient that conflict with the committee's average gradient, using the local elasticity of neural networks to limit decision-boundary drift.
- Introduced a periodically refreshed committee with cached (stale) gradients to cut per-step overhead; improved accuracy by **3.5%** over standard SGD on CIFAR-100 (VGG16), and found the gains do not carry over to residual architectures (ResNet18/20).

Selected publications

*359 total citations, h-index 5 (Google Scholar, Oct 2026). *denotes equal contribution.*

1. Richeek Das*, **Sahasra Ranjan***, Shreya Pathak, Preethi Jyothi. *Improving Pretraining Techniques for Code-Switched NLP.* **ACL 2023** Outstanding Paper Award
2. Alexander Kroll, **Sahasra Ranjan**, Martin K. M. Engqvist, Martin J. Lercher. *A general model to predict small molecule substrates of enzymes based on machine and deep learning.* **Nature Communications, 2023** *(265 citations)*
3. Alexander Kroll, **Sahasra Ranjan**, Martin J. Lercher. *A multimodal Transformer network for protein-small molecule interactions enhances predictions of kinase inhibition and enzyme-substrate relationships.* **PLoS Computational Biology, 2024** *(48 citations)*

Technical skills

Languages: Python, C++, Go, TypeScript, Swift, SQL
Machine Learning: PyTorch, JAX, Hugging Face Transformers, vLLM, protein language models
AI Systems: agent evaluation (LLM-as-judge, trace replay), synthetic data generation, agent memory, RAG, model routing
Real-time & Infrastructure: live audio and vision models, WebRTC, SIP, LiveKit, gRPC, Docker, Kubernetes, GCP, PostgreSQL

Teaching

Teaching Assistant, CS725: Foundations of Machine Learning (Prof. Preethi Jyothi), IIT Bombay *Fall 2022*
Teaching Assistant, CS753: Automatic Speech Recognition (Prof. Preethi Jyothi), IIT Bombay *Spring 2023*